



哈爾濱工業大學
HARBIN INSTITUTE OF TECHNOLOGY

“OrderCore” 设计报告

——第十届“龙芯杯”全国大学生计算机系统能力大赛



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设计概述

- » 非对称双发射，前后端解耦设计
- » 8KB icache
- » 8KB L1-dcache + 64KB L2-dcache
- » 分支预测正确率 $\approx 94\%$
- » 支持通过uboot引导启动linux
- » 丰富的外设支持（LCD、VGA、I2S…）

» 性能成绩：

» 主频106MHz

» 相比于baseline
openla500

整体加速比：5.15

IPC加速比：1.59

实际IPC：1.16



性能表现

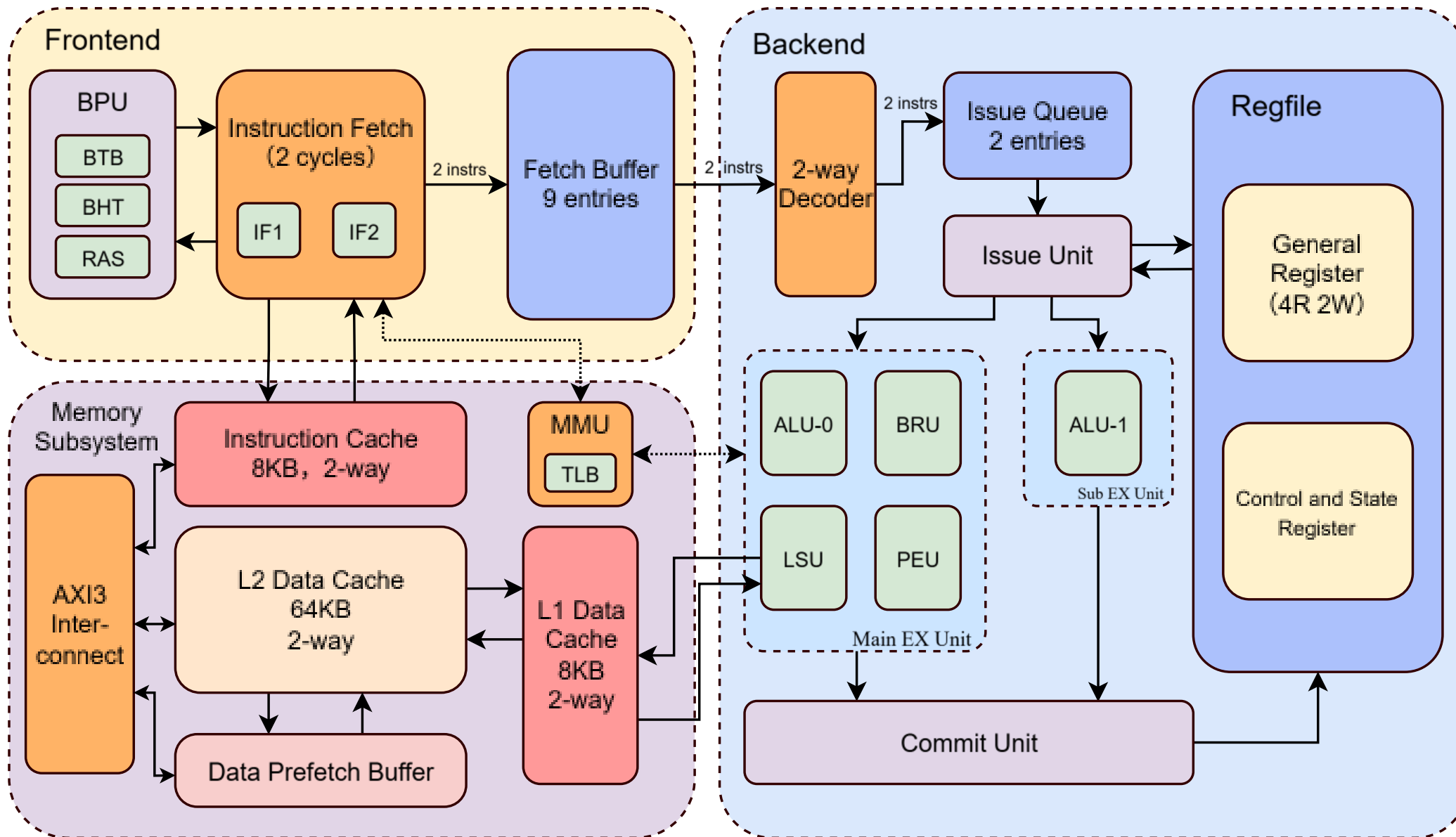
» 主频106MHz

Benchmark	IPCup	Benchmark	IPCup
bitcount	1.63	fireye_A0	7.46
bubble_sort	1.01	fireye_B2	1.50
coremark	1.20	fireye_C0	1.01
crc32	1.20	fireye_D1	2.38
dhrystone	1.14	fireye_I2	1.10
quick_sort	1.23	inner_product	3.25
select_sort	1.33	lookup_table	1.87
sha	1.20	loop_induction	2.20
stream_copy	1.30	my_memcmp	2.11
stringsearch	1.47	Minmax_sequence	1.30
Geo.Mean		1.59	





微架构



前端-分支预测器

» 局部历史预测器 (BHT+PHT)

1024项2位BHR, 2048项PHR

索引 = $PC[21:12] \wedge PC[11:2]$, 无tag比对

» 分支目标缓冲区 (BTB)

2-way组相联, 每路1024条目

索引 = $PC[11:2]$, 标记位 = $PC[19:12]$

» 返回地址堆栈 (RAS)

最大深度为8

1024位置信位与标记位

预测跳转 = $(BHT_hit \mid RAS_hit) \& BTB_hit$

Benchmark	Hit Rate	Benchmark	Hit Rate
bitcount	97.7%	fireye_A0	99.5%
bubble_sort	83.9%	fireye_B2	95.6%
coremark	93.1%	fireye_C0	91.8%
crc32	99.6%	fireye_D1	97.0%
dhrystone	99.5%	fireye_I2	96.8%
quick_sort	84.4%	inner_product	99.9%
select_sort	97.1%	lookup_table	99.4%
sha	99.4%	loop_induction	61.6%
stream_copy	99.2%	my_memcmp	99.9%
stringsearch	93.8%	Minmax_sequence	98.0%
Geo.Mean		93.9%	



前端-指令缓冲区

» FIFO先进先出队列

物理深度为16项，实际使用最大深度为9项

» 作用一

减少流水线控制信号反压，ID_allowin仅由fetch_buffer是否有空项决定

» 作用二

解耦取指与后端执行，在流水线后端暂停时提前取指，掩盖取指访存延迟



后端设计

» 发射/写回宽度均为2

» 允许适当对调的双发射机制

两条指令4种搭配组合中，仅一种情况不允许同时发射

指令1 \ 指令2	normal	abnormal
normal	√	√
abnormal	√	×

» 高效前递路径

ALU指令: EX->IS 、 WB->IS

所有指令: WB->IS

```
1  assign inst_normal = inst_add_w | inst_sub_w | inst_addi_w | inst_lu12i_w |  
2                                inst_slt | inst_sltu | inst_slti | inst_sltui |  
3                                inst_pcaddu12i | inst_and | inst_or | inst_nor | inst_xor |  
4                                inst_andi | inst_ori | inst_xori |  
5                                inst_mul_w | inst_mulh_w | inst_mulh_wu |  
6                                inst_sll_w | inst_srl_w | inst_sra_w |  
7                                inst_slli_w | inst_srli_w | inst_srai_w |
```



Cache

» 指令cache

8KB, 2-way, 组数64, 行大小64B, LRU替换算法

两个读端口, 一套hit逻辑, 节约资源

» 数据Cache

L1 8KB, 2-way, 组数64, 行大小64B, LRU替换算法

两周期流水访问

L2 64KB, 2-way, 组数512, 行大小64B, LRU替换算法

四周期访问

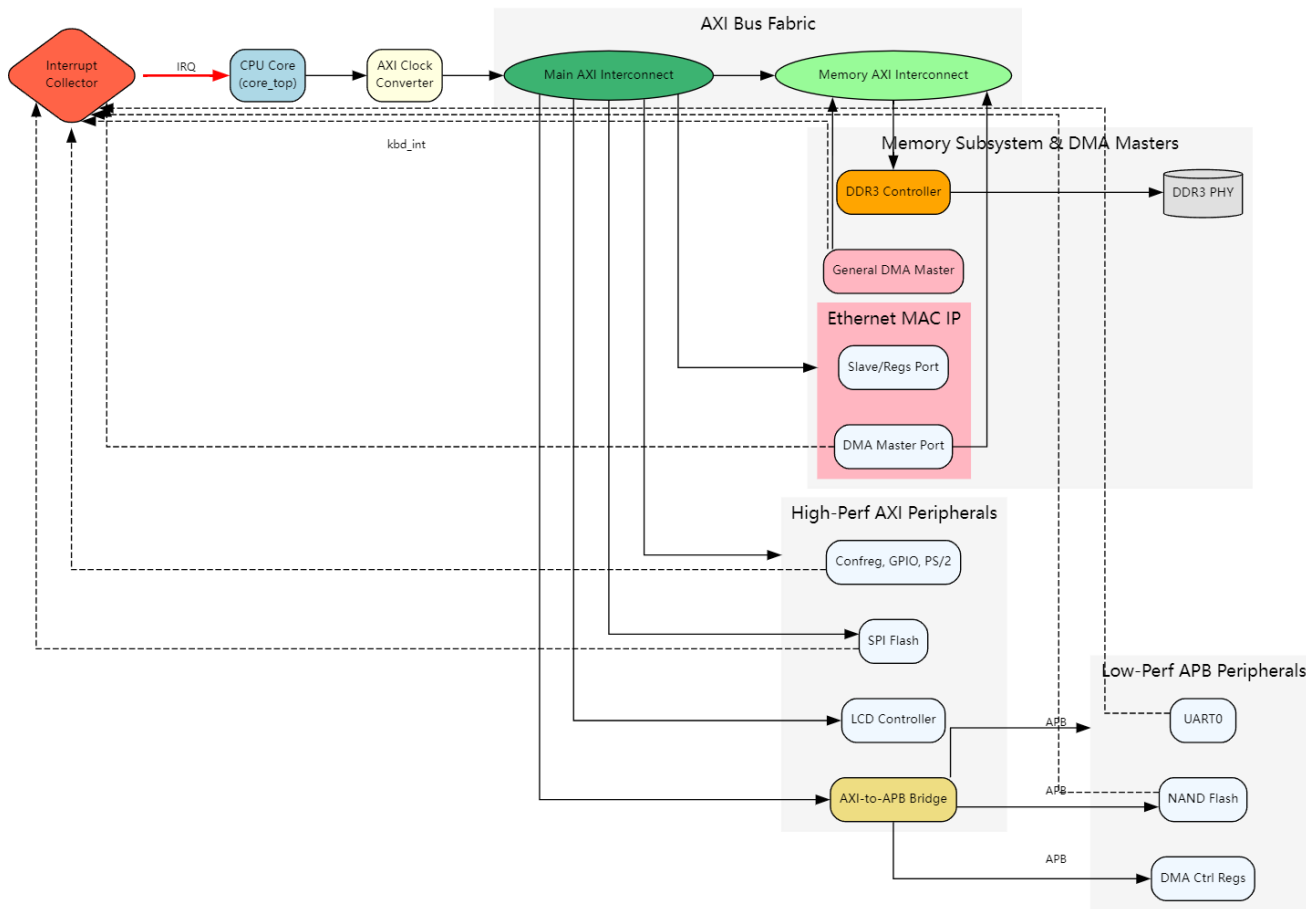
» 硬件预取机制



SoC设计

- » 支持除USB外的所有板载外设输出
- » 利用拓展IO实现I2S音频输出支持
- » 本SoC支持驱动的外设共有：

UART串口、GPIO（驱动LED灯与数码管）、PS/2、SPI Flash、8080 TFT LCD、VGA、I2S、NAND Flash、DMA控制器和PHY以太网设备



SoC演示

```
serial-com7 (1) x
u-boot@LoongsonSoC# ps2test
PS/2 Keyboard Test - Press keys (ESC to exit)
Mode: Single-key

[key 0x00000016]
Status Reg: 0x00000001
Data Reg: 0x16000000
Decoded: '1' (0x31)

-----

[key 0x0000001c]
Status Reg: 0x00000001
Data Reg: 0x1c000000
Decoded: 'a' (0x61)

-----

[key 0x00000032]
Status Reg: 0x00000001
Data Reg: 0x32000000
Decoded: 'b' (0x62)

-----

[key 0x00000021]
Status Reg: 0x00000001
Data Reg: 0x21000000
Decoded: 'c' (0x63)

-----

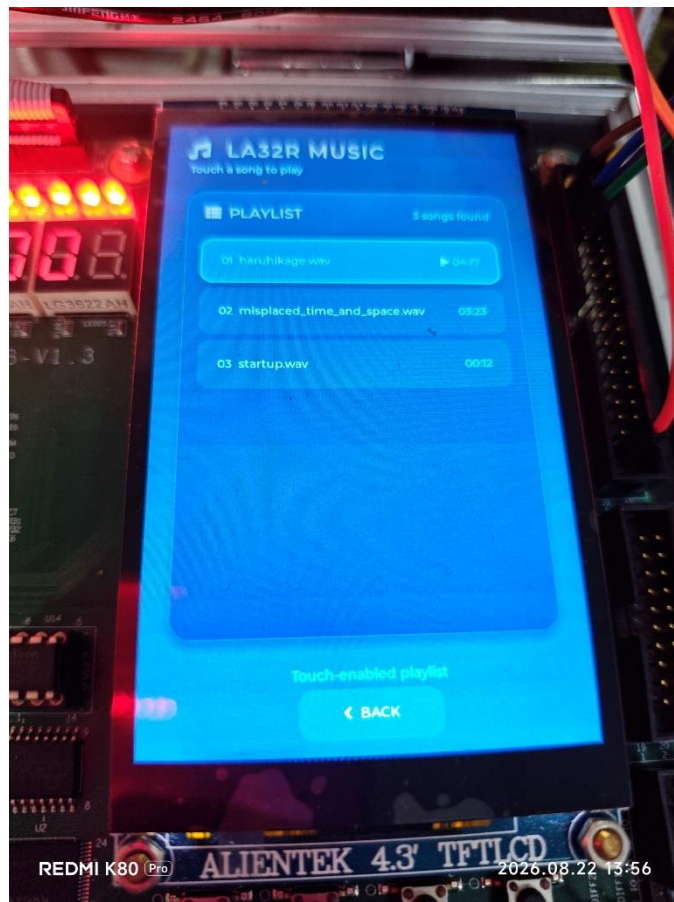
[key 0x00000016]
Status Reg: 0x00000001
Data Reg: 0x16000000
Decoded: '1' (0x31)

-----

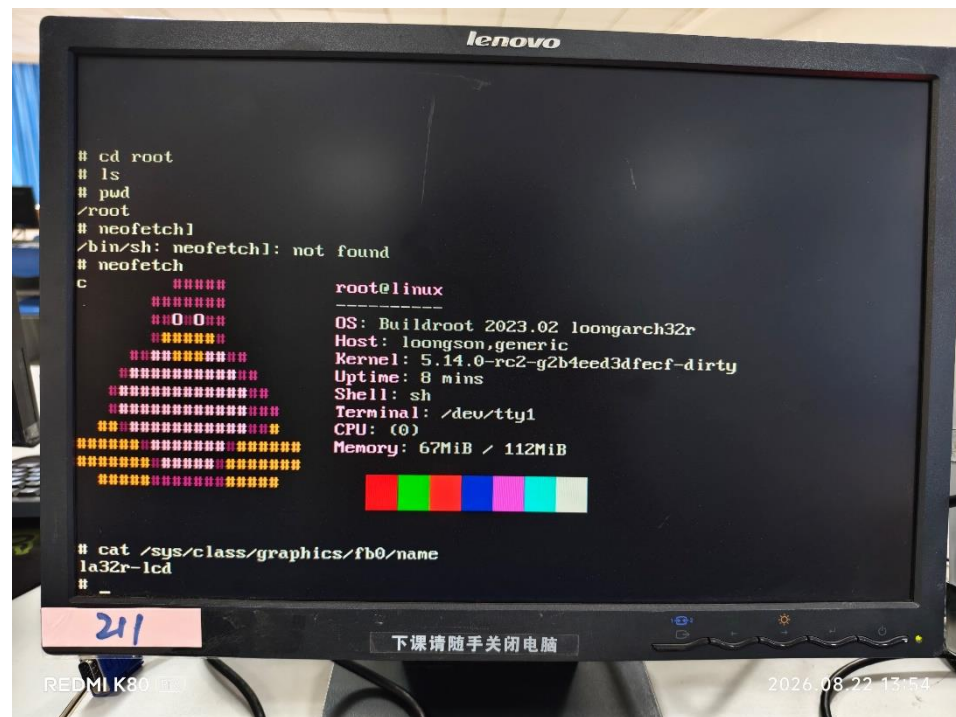
[key 0x00000076]
Status Reg: 0x00000001
Data Reg: 0x76000000
Decoded: Special key (0x76)

PS/2 Test Complete
```

» PS/2键盘测试



» LCD测试



» VGA测试



操作系统

» linux

自行构建busybox作为根文件系统的linux内核。成功通过uboot引导启动linux并运行ls命令。

添加对neofetch等多条指令的支持；新增LCD、VGA、I2S等驱动支持；新增con2fbmap命令，支持在系统启动后动态切换虚拟终端的输出设备（VGA/LCD）

```
serial-com6 - SecureCRT
File Edit View Options Transfer Script Tools Window Help
Enter host <Alt+R>
serial-com6 x
4.384000 i2c /dev entries driver
4.448000 Goodix-TS 0-0014: ID 1158, version: 0100
4.544000 input: Goodix Capacitive TouchScreen as /devices/platform/soc/1fa10000.i2c/i2c-0/0-0014/input/input1
4.664000 IR MCE Keyboard/mouse protocol handler initialized
4.720000 hid: raw HID events driver (C) Jiri Kosina
4.824000 NET: Registered PF_INET6 protocol family
4.896000 Segment Routing with IPv6
4.952000 sit: IPv6, IPv4 and MPLS over IPv4 tunneling driver
5.016000 sctp: Hash tables configured (bind 1024/1024)
5.096000 Btrfs loaded, crc32c=crc32c-generic, zoned=yes
9.312000 key type encrypted registered
9.340000 ALSA device list:
9.368000 #0: LA32R MAX98357A
9.536000 Freeing unused kernel image (initmem) memory: 7756K
9.564000 This architecture does not have kernel memory protection.
9.592000 Run /init as init process
Starting syslogd: OK
Starting klogd: OK
Running sysctl: OK
Saving 2048 bits of non-creditable seed for next boot
Starting system message bus: [ 10.984000] random: dbus-uuidgen: uninitialized urandom read (12 bytes read)
[ 11.016000] random: dbus-uuidgen: uninitialized urandom read (8 bytes read)
[ 11.088000] random: dbus-daemon: uninitialized urandom read (12 bytes read)
done
Starting rpcbind: OK
Starting network: OK
Starting NFS statd: OK
Starting NFS services: OK
Starting NFS daemon: rpc.nfsd: unable to access /proc/fs/nfsd errno 2 (No such file or directory).
Please try, as root, 'mount -t nfsd nfsd /proc/fs/nfsd' and then restart rpc.nfsd to correct the problem
FAIL
Starting NFS mountd: OK

Welcome to Linux
linux login: root
Password:
# ls /
bin      init     linuxrc  opt      run      tmp
dev      lib      media   proc     sbin     usr
etc      lib32   mnt     root     sys      var
```

» 成功启动linux



设计总结

- » 较高性能的短流水顺序双发射处理器
- » 平衡设计复杂度与性能
- » 通过高效双发机制和前递网络，以较小面积代价换取较大提升

感谢各位老师的倾听，请批评指正！





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Thanks!